BPM2_ESOFT QUIZ 1_S2026_GD

**# of Questions:** 30

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## **Question #:** 1

A 25-year-old woman comes to the physician because of a 3-day history of pain in her external genitalia. Physical examination shows redness and swelling of the labia majora. She was diagnosed with an infected ingrown hair on the labia majora. Which of the following groups of lymph nodes will initially drain the infected area?

A. Lumbar
B. Deep inguinal
C. Internal iliac
✓D. Superficial inguinal
E. Axillary

**Rationale:** The superficial inguinal nodes drain the skin of the perineum, including the labia majora in females. The exceptions are the glans clitoris and labia minora, which are drained by the deep inguinal nodes.

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## **Question #:** 2

A 25-year-old man comes to the emergency department because of a 2-day history of severe pelvic pain associated with fever and chills. His temperature is 38°C (100.4°F), pulse is 95/mins, blood pressure is 90/70mm Hg and respirations are 22/min. Imaging studies show an abscess between the perineal membrane and the fascia of the pelvic diaphragm. Which of the following structures is most likely located within this perineal space?

A. Perineal body
✓B. Bulbourethral glands

C. Superficial transverse perineal muscle
D. Ischiocavernosus muscle
E. Bulbospongiosus muscle

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**Question #:** 3

A 17-year-old boy is brought to the emergency department because of discoloration and swelling in his penis, scrotum and anterior abdominal wall, He was riding his bicycle when he went over a depression in the road causing him to fall, straddling the handle of his bicycle. His symptoms are characteristic of extravasation of urine. Which of the following structures is most likely damaged?

A. Ureter
B. Urinary bladder
C. Prostatic urethra
D. Membranous urethra
✓E. Spongy urethra

**Rationale:** The patient has an anterior urethral tear, with damage to the spongy urethra and to bucks fascia. This results in extravasation of urine and blood around the penis, into the scrotum, the superficial perineal pouch and continuing to the anterior abdominal wall.

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**Question #:** 4

A 75-year-old man comes to the physician because of difficulty urinating and urinary hesitancy for the past 3 weeks. He notes that he usually 'starts and stops' when urinating and never completely empties the bladder. Cystourethrogram shows a narrowing of the urethra due to an enlarged prostate. Which of the following areas of the prostate is most likely enlarged, producing the symptoms in this patient?

A. Central zone

B. Peripheral zone
C. Anterior stroma
✓D. Periurethral zone

**Rationale:** The periurethral zone is directly adjacent to the prostatic urethra. As a result, enlargement of this zone will cause narrowing of the urethra and result in the patient's symptoms.

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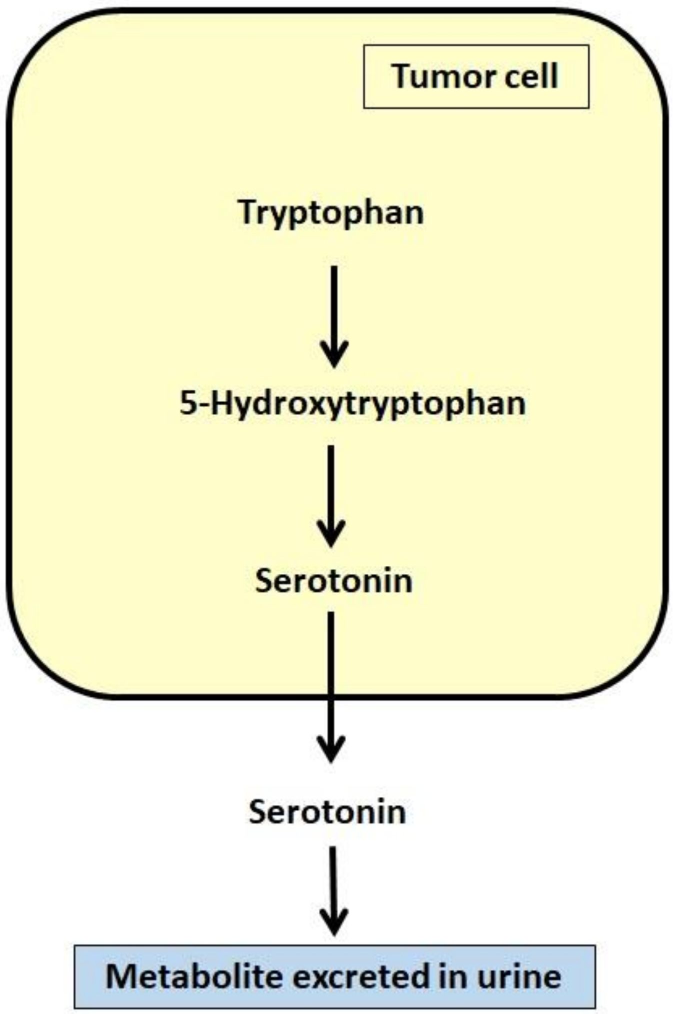
**Question #:** 5

A 38-year-old woman comes to the physician because of a 2-month history of episodes of flushing, abdominal pain, and diarrhea. An abdominal CT scan shows a tumor in the jejunum with involvement of the mesenteric lymph nodes. Biopsy confirms a neuroendocrine tumor of the jejunum. Urinalysis shows elevated levels of a specific metabolite. The metabolism in a typical tumor cell is shown in the figure. Which of the following enzymes is most likely required for the formation of the urinary metabolite?

✓A. Monoamine oxidase
B. 5-hydroxytryptophan decarboxylase
C. Catechol O-methyl transferase
D. Tryptophan hydroxylase
E. DOPA decarboxylase

**Rationale:** The patient has a neuroendocrine tumor or a Carcinoid tumor with overproduction of serotonin (5-hydroxytryptamine).
Serotonin is converted by Monoamine oxidase (MAO) to 5-hydroxyindoleacetic acid (5HIAA). Urinalysis shows elevated levels of 5-HIAA.
Metanephrine and vanillyl mandelic acid are formed from epinephrine and norepinephrine metabolism. Increased urinary VMA and metanephrine are found in Pheochromocytoma. Metanephrine is formed by the action of Catechol-O-methyl transferase (COMT) on epinephrine and norepinephrine. VMA is formed from metanephrine by the action of MAO.
Dopamine forms homovanillic acid (HVA) by COMT and MAO.
5-hydroxytryptophan is the precursor of serotonin. It is converted to serotonin by a 5-hydroxytryptophan decarboxylase requiring PLP (vitamin B6).
Tryptophan is converted to 5-hydroxytryptophan by tryptophan hydroxylase, which requires tetrahydrobiopterin as a coenzyme.

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**Question #:** 6

A group of investigators is studying steroid metabolism in peripheral tissues in postmenopausal females. They incubate radioactively labeled androstenedione, a hormone released by the adrenal cortex, with adipocytes to measure the amount of estrogen produced from this androgen. Which of the following enzymes is most likely present in adipocytes and is responsible for estrogen formation from this androgen?

A. CYP 11A
B. CYP 11B

C. CYP 17
✓D. CYP 19
E. CYP 21
F. Desmolase
G. 3-beta hydroxysteroid dehydrogenase

**Rationale:** CYP 19 (also known as Aromatase) is found in adipocytes. It mainly uses androstenedione for estrogen synthesis.

5α-reductase converts testosterone → dihydrotestosterone (DHT)

17α-hydroxylase, 21-hydroxylase, and 11β-hydroxylase are involved in adrenal steroid synthesis (cortisol, aldosterone, and androgen precursors)

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**Question #:** 7

A 45-year-old woman comes to the physician for advice regarding weight management. She has a family history of obesity. Her waist circumference is 75 cm (29.5 in), and her BMI is 25 kg/m². Laboratory studies show a high insulin-to-glucagon ratio, and she has no clinical or laboratory features of insulin resistance. Which of the following is most likely observed in this metabolic state?

A. Increased skeletal muscle protein degradation
✓B. Increased expression of lipoprotein lipase
C. Reduced synthesis of triglycerides in adipocytes
D. Decreased GLUT-4 in muscle
E. Increased glucose output by liver
F. Increased blood glucose levels
G. Decreased glucose uptake by liver

**Rationale:** The woman is a normal patient with anthropometry and weight within normal limits. Lipoprotein lipase activity is induced by insulin; All the other metabolic changes are seen in patients with insulin resistance and those with type 1 and type 2 diabetes mellitus.

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**Question #:** 8

A 75-year-old man is brought to the physician by his daughter because of a 1-year history of progressive memory loss. His daughter reports that he has been getting lost in familiar places, has difficulty paying bills, and requires assistance with meal preparation and personal hygiene. His medications and nutrient supplements are closely monitored. Over-supplementation of which of the following nutrients is best avoided in this patient?

A. Vitamin C
B. Protein
C. Folic acid
✓D. Vitamin A
E. Vitamin D
F. Calcium

**Rationale:** Over-supplementation of vitamin A has the risk of developing hypervitaminosis A in older people.
Calcium and vitamin D supplements are generally recommended due to poor absorption and the higher risk of osteoporosis.

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**Question #:** 9

A 10-day-old boy is brought to the physician by his parents because of a 2-day history of poor feeding, vomiting, and failure to thrive. Laboratory studies show:

- Hyponatremia

- Hyperkalemia

Chromosomal analysis shows 46, XY. Further laboratory studies confirm a diagnosis of congenital adrenal hyperplasia (CAH) due to an inherited enzyme deficiency. Which of the following reactions is most likely catalyzed by the deficient enzyme in this patient?

✓A. Conversion of progesterone to 11-deoxycorticosterone
B. Conversion of testosterone to estradiol
C. Conversion of cholesterol to pregnenolone
D. Conversion of pregnenolone to progesterone
E. Conversion of androstenedione to testosterone
F. Conversion of 11-deoxycorticosterone to corticosterone
G. Conversion of progesterone to 17-hydroxyprogesterone

**Rationale:** Hyponatremia and hyperkalemia with low blood pressure suggest CAH due to deficiency of CYP21 (21-hydroxylase).
Note that 21-hydroxylase is required for the synthesis of both the mineralocorticoids and glucocorticoids. It converts 17-hydroxyprogesterone to 11-deoxycortisol which is next converted to cortisol by 11-beta hydroxylase. 21-hydroxylase is also required for the conversion of progesterone to 11-deoxycorticosterone which is later converted to corticosterone and then to aldosterone (mineralocorticoid).
Review the enzymes catalyzing the other listed reactions.

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**Question #:** 10

A 16-year-old boy is brought to the emergency department in a confused and disoriented state. His heart rate is elevated. Laboratory studies reveal a plasma glucose of 50 mg/dL (reference range: 70-100 mg/dL). His mother has type 2 diabetes and takes sulfonylurea therapy. A provisional diagnosis of factitious hypoglycemia due to sulfonylurea ingestion is considered. Which of the following laboratory findings is most likely to confirm this diagnosis?

A. Reduced serum insulin and proinsulin levels
B. Elevated serum C-peptide and reduced insulin levels
C. Elevated serum insulin and low C-peptide levels
✓D. Elevated serum proinsulin and C-peptide levels
E. Reduced serum insulin and C-peptide levels

**Rationale:** Sulfonylurea stimulates the beta-cells of the pancreas to secrete insulin. Serum proinsulin and C-peptide levels are indicators of endogenous insulin production and are raised in patients with factitious hypoglycemia due to sulfonylurea overdose. Sulfonylurea will be present in serum.

In exogenous insulin overdose, serum insulin concentration is elevated, whereas proinsulin and C-peptide concentrations are very low. Exogenous insulin inhibits the beta-cell insulin release.
In insulinoma, serum insulin, proinsulin, and C-peptide concentrations are elevated.

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**Question #:** 11

An 85-year-old man comes to the physician for a routine health examination. The physician has been following him for the past 20 years and notes that his weight has remained stable. Laboratory studies are within normal limits, and he is in good health. Which of the following metabolic changes is most likely observed in this patient?

✓A. Negative nitrogen balance
B. Increased cellular oxidant defense mechanisms
C. Increased caloric requirement
D. Decreased rate of muscle protein catabolism
E. Increased intestinal absorption of nutrients

**Rationale:** Sarcopenia (loss of muscle or lean body mass) is a metabolic change associated with aging. There is decreased oxidant defense mechanisms, decreased caloric requirement, decreased BMR, increased muscle protein breakdown, and decreased absorption of nutrients related to aging.

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**Question #:** 12

A 2-week-old girl is brought to the physician by her parents because of a 1-day history of persistent vomiting. The childbirth took place at home. Her blood pressure is low, and physical examination shows ambiguous genitalia and signs of dehydration. Laboratory studies show:

- Serum glucose: 55 mg/dL (Normal: 70-100 mg/dL)

- Serum Sodium: 107 mmol/L (normal: 135–145 mmol/L)

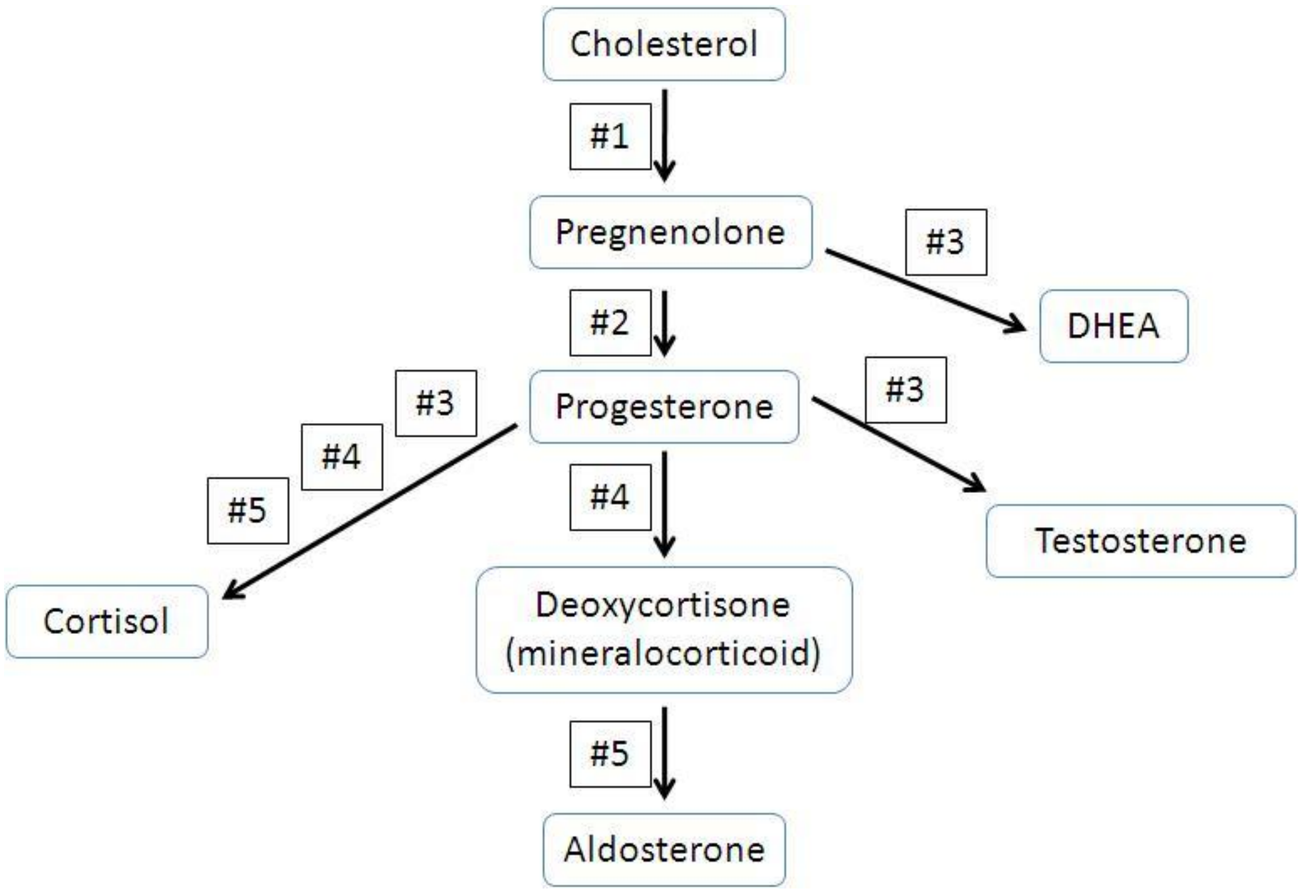
- Serum Potassium: 6.5 mmol/L (normal 3.5–5.0 mmol/L)

An inherited enzyme deficiency in the adrenal steroidogenesis pathway is confirmed. Which of the following numbered enzymes in the pathway shown is most likely to be deficient in this patient?

A. Enzyme #1
B. Enzyme #2
C. Enzyme #3
✓D. Enzyme #4
E. Enzyme #5

**Rationale:** The vignette describes a patient with CAH due to CYP-21 deficiency, Enzyme#4 in the figure. Note the hypoglycemia, hyponatremia, hypotension, and hyperkalemia.
Key features include:
Salt-wasting: Low aldosterone → hyponatremia, hyperkalemia, hypotension
Cortisol deficiency: Hypoglycemia
Androgen excess: Ambiguous genitalia in genetic females; Elevated DHEAS
Onset: Usually first 1–2 weeks of life in the salt-wasting form

Enzyme#1: Cholesterol side chain cleavage enzyme or Desmolase or CYP11A
Enzyme#2: 3-beta-hydroxysteroid dehydrogenase; Rare, ambiguous genitalia in both sexes; mild salt-wasting
Enzyme#3: 17-hydroxylase (CYP17); Hypertension, hypokalemia, sexual infantilism; no virilization
Enzyme#4: 21-hydroxylase (CYP21)
Enzyme#5: 11-beta hydroxylase (CYP11B1); Hypertension, virilization; mild salt retention

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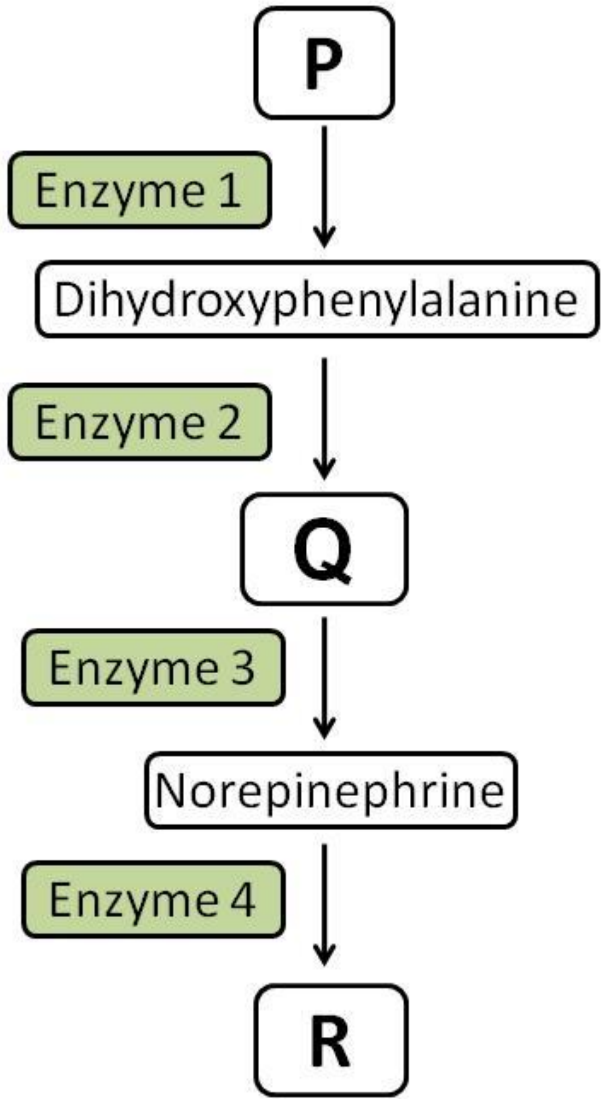
**Question #:** 13

An investigator is studying a metabolic pathway involved in the synthesis of an important family of biogenic amines. A partially labeled figure of the pathway is shown, depicting sequential enzymatic steps converting amino acid precursors into active neurotransmitters. Which of the following statements best characterizes this pathway?

A. Enzyme 1 is tryptophan hydroxylase
B. Enzyme 1 requires vitamin C as cofactor
C. Enzyme 2 forms L-DOPA
D. Enzyme 2 requires tetrahydrobiopterin (BH4) as a cofactor
E. Enzyme 3 catalyzes a ring hydroxylation reaction
F. Enzyme 3 requires vitamin B6 as a coenzyme
✓G. Enzyme 4 catalyzes a methylation reaction
H. Enzyme 4 forms vanillylmandelic acid (VMA)

**Rationale:** P - is Tyrosine; Q - is Dopamine, and R - is Epinephrine. Enzyme 1 is Tyrosine hydroxylase, which requires tetrahydrobiopterin as a coenzyme. Enzyme 2 is DOPA decarboxylase, which requires PLP as a coenzyme; Enzyme 3 is Dopamine beta hydroxylase, which requires vitamin C as a coenzyme. Enzyme 4 is phenylethanolamine N-methyl transferase (PNMT), which requires SAM as a methyl donor.

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**Question #:** 14

A 40-year-old woman comes to the physician because of a 1-month history of progressive dizziness when she gets out of bed in the morning. She also reports episodes of flushing, warmth, and sweating. Laboratory testing confirms a diagnosis of carcinoid syndrome. Which of the following urinary metabolites is most likely increased in this patient?

A. Vanillylmandelic acid (VMA)
✓B. 5-hydroxyindoleacetic acid (5-HIAA)
C. 17-ketosteroids
D. Homovanillic acid (HVA)
E. Epinephrine
F. r-T3 (reverse T3)
G. Metanephrine

**Rationale:** 5HIAA is formed by the action of MAO on Serotonin (5-HT).
VMA and metanephrine are the degradation products of catecholamines (epinephrine and norepinephrine). Urinary metanephrine and VMA are increased in pheochromocytoma.
HVA is the degradation product of dopamine metabolism.
The degradation of thyroid hormone forms reverse T3.

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**Question #:** 15

A 25-year-old man comes to the physician because of a 2-week history of headaches, dizziness and general weakness and a 1-day history of nausea and vomiting. His blood pressure is 120/70 mm Hg and pulse is 70/min. Physical examination shows an abnormal accumulation of fat at the posterior aspect of the neck. Laboratory studies of the blood show glucose levels of 430 mg/dL (normal range 70-110mg/dL). An abdominal CT-scan shows a tumor involving the adrenal gland. Which of the following hormones produced by the adrenal gland is most likely to be the direct cause of the sustained elevated glucose levels seen in this patient?

A. Aldosterone
B. Androgen
C. Norepinephrine
D. Erythropoietin
✓E. Cortisol

**Rationale:** The cells of the zona fasciculata secrete the glucocorticoid cortisol.

Cortisol functions in the regulation of blood glucose as well as different forms of metabolism. Therefore, an increased level of serum cortisol can affect glucose homeostasis and result in hyperglycemia, as observed in this case.
Aldosterone is produced in cortex of the adrenal gland as well, but is involved in blood pressure homeostasis as well as electrolyte and water balance.
Abnormally elevated levels of androgen will result excessive hair growths and masculinization. The zona reticularis of the adrenal gland cortex will be affected with these physical symptoms. Norepinephrine is part of the sympathetic response. While it can elevate blood glucose levels, this is generally the case whenever the body is enduring stressful situations. Erythropoietin aids in the production of RBC during hypoxia.

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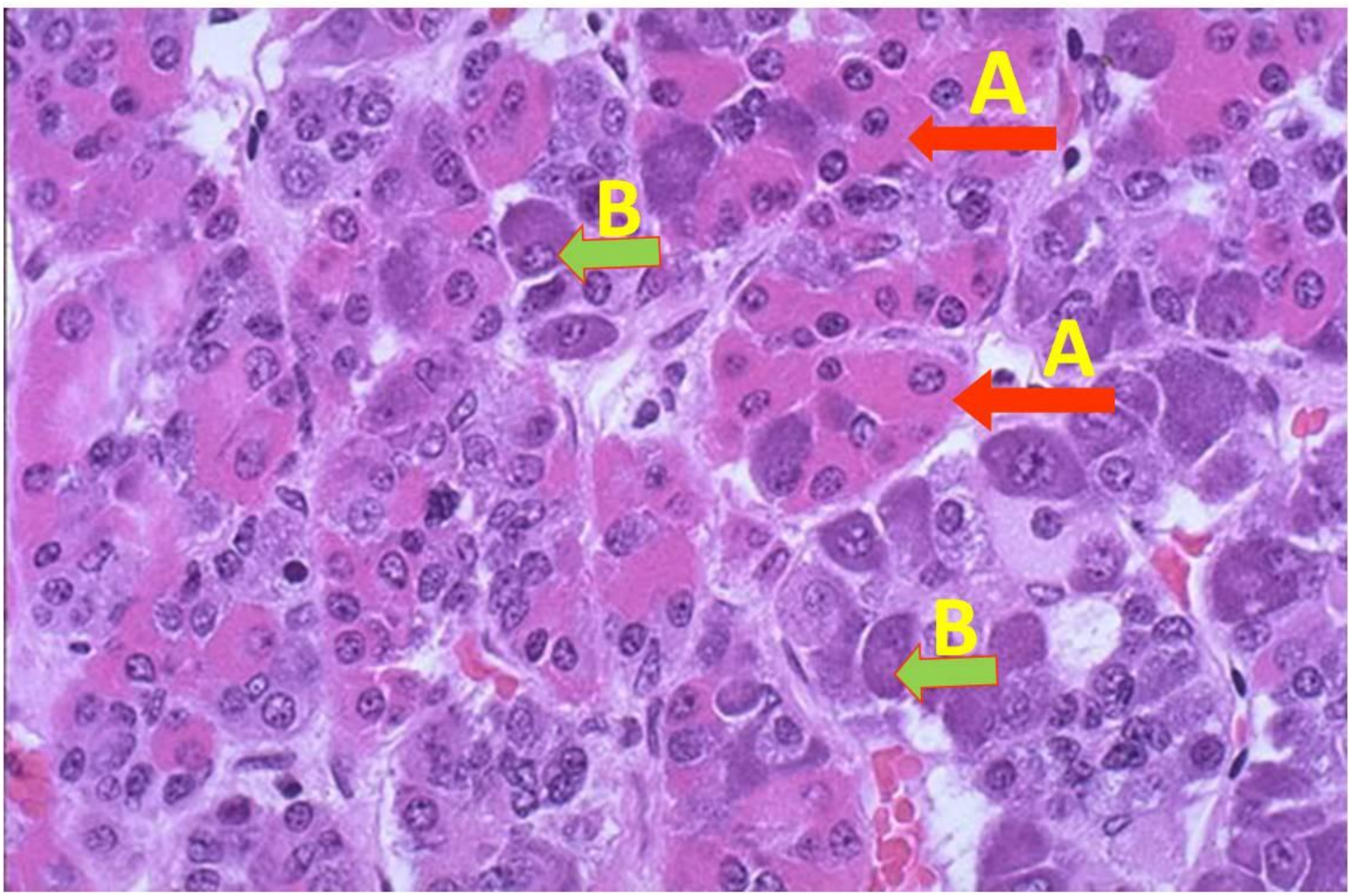
**Question #:** 16

A 31-year-old woman comes to the physician because of a 2-month history of headaches, weight loss of 3.17-kg (7-lb), increased sweating, increased frequency of stool, and hot flushes. Physical examination shows a wasted young female with bulging eyes and an irregular heart beat. Laboratory studies show elevated TSH, T3 and T4 levels. CT-scan of the head shows a pituitary adenoma. She is diagnosed with a tumor of the cells labelled at the tip of the **pointer "B"** in the attached photomicrograph. Which of the following cell type is most likely affected?

A. Chromophobes
B. Spongiocytes
✓C. Thyrotropes
D. Somatotropes
E. Lactotropes

**Rationale:** Thyrotropes are a subset of the basophils that secrete thyroid-stimulating hormone. The patient most likely has a thyrotrope adenoma, based on her symptoms.
Spongiocytes are found in the cortex of the adrenal gland, specifically in the zona fasiculata. There they produce cortisol.
Somatotropes and lactotropes are examples of acidophils, and produce growth hormone and prolactin respectively. Acidophils are identified by arrow "A" in this

image.
Chromophobes represent the other cell population found in the pars distalis of the anterior pituitary whose function is largely unknown.

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**Question #:** 17

A 48-year-old man was brought to the emergency department because of sudden onset of dizziness, sweating and confusion. His blood pressure is 122/60 mm Hg, pulse is 92/min, respirations are 24/min and blood glucose is 20 mg/dL (normal is 70 - 110 mg/dL). An abdominal MRI shows a mass in his pancreas. Microscopic examination of a biopsy specimen of the pancreatic mass shows an insulinoma. Which of the following features best characterizes the pancreatic cell population most likely affected in this condition?

A. Stimulated by a hormone stored in the supraoptic nucleus
B. Inhibited by dopamine produced by the hypothalamus
C. Mostly found in the periphery of the islets of Langerhan
D. Modified post-ganglionic sympathetic neurons
✓E. Generally located in the central core of the islets of Langerhan

**Rationale:** The beta cells of the pancreas produce and secrete insulin. Insulin controls blood glucose levels by facilitating the uptake of glucose from the blood into the myocytes, hepatocytes and adipocytes. An insulinoma is an insulin producing tumor. Therefore the beta cells are directly affected in these kinds of tumors. Beta cells contribute to 70% of the islets and are generally concentrated in the core of these islets. Alpha cells are found predominantly at the periphery. Post-ganglionic sympathetic neurons are found in the adrenal medulla. The supraoptic nucleus produces ADH. ADH acts on the collecting ducts of the kidneys to promote water reabsorption. Dopamine inhibits the secretion of the hormone prolactin. Prolactin is elevated in the tumor prolactinoma, and not insulinoma.

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**Question #:** 18

A 37-year-old man comes to the physician because of a 5-month history of fatigue, low energy, hoarseness, and a 5-kg (11-lb) weight gain. Physical examination shows non-pitting edema of the extremities, sparse body hair, bradycardia, and sluggish deep tendon reflexes. Laboratory studies show low serum concentrations of thyroid-stimulating hormone (TSH), Triiodothyronine (T3), and Thyroxine (T4). Which of the following is the most likely diagnosis?

A. Graves disease
B. Hashimoto thyroiditis
✓C. Secondary hypothyroidism
D. Primary hypothyroidism
E. Iodine deficiency

**Rationale:** This patient has low concentrations of both TSH and thyroid hormones. Therefore, he has a pituitary defect. This is called secondary hypothyroidism.

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**Question #:** 19

A 68-year-old man is brought to the physician because of a 4-day history of headache, difficulty concentrating, muscle weakness, and decreased frequency of urination. He was diagnosed with lung cancer 3 months ago and is on chemotherapy. His temperature is 37°C (98.6°F), pulse is 78/min and respirations are 18/min. Physical examination shows no abnormalities. Mental status examination shows disorientation to place and time, and difficulty following verbal commands. Laboratory studies show a serum sodium of 126 mEq/L (Normal: 135-145), serum osmolality of 274 mOsmol/kg $H_2O$ (Normal: 290 mOsmol/kg $H_2O$), and elevated serum vasopressin. He is diagnosed with syndrome of inappropriate ADH (SIADH). Which of the following physical findings is most likely in this patient?

A. Increased blood pressure
✓B. Normal blood pressure
C. Dry mucous membranes
D. Yellow sclera
E. Loud second heart sound

**Rationale:** This patient has excess secretion of vasopressin due to a lung cancer. This patient retains large volume of water because of action of vasopressin which results in hyponatremia and low serum osmolality. Decreased serum osmolality makes water movement into the intracellular compartment and swelling of cells in organs. Such change in size of cells in the brain has resulted in change in mental status of this patient. Although water retention takes place, there is no expansion of the intravascular volume (plasma volume). Therefore, the blood volume and blood pressure remain normal in this patient.

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**Question #:** 20

A 28-year-old woman, at 39 weeks' gestation comes to the emergency department because of a 1-hour history of labor pain. Her pregnancy has been complicated with gestational diabetes, but she has had regular antenatal care. Her vital signs are within normal limits. Physical examination shows intermittent uterine contractions and normal fetal heart sounds. Ultrasonography shows a slightly bigger size of the fetus because of her medical condition. In order to avoid delivery-related complications, the physician decides to induce labor by administering an intravenous oxytocin drip. Which of the following most likely is an additional physiologic effect of oxytocin?

A. Milk production by the mammary gland
B. Contraction of the vascular smooth muscles
C. Inhibition of prolactin secretion
✓D. Contraction of the myoepithelial cells in the mammary gland
E. Inhibition of the menstrual cycle

**Rationale:** Oxytocin has 2 major physiologic actions: a) Stimulating uterine contraction, and b) stimulating myoepithelial cells surrounding the alveoli of the mammary gland milk ejection. Milk production is the action of prolactin.

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**Question #:** 21

A 24-year-old man is brought to the physician by his father because of a 3-month history of decreased facial and body hairs and enlargement of breasts. The father says his son rarely shaves his face. He was diagnosed with Schizophrenia 4 months ago and is taking anti-psychotics. His vital signs are within normal limits. Physical examination shows bilateral gynecomastia and scanty facial, axillary, pubic and body hairs. Laboratory studies show elevated plasma prolactin level. Which of the following mechanisms best explains the laboratory finding observed in this patient?

A. Stimulation of lactotrophs by TRH
B. Damage to the pituitary stalk
C. Stimulation of lactotrophs by ACTH
✓D. Inhibition of the hypothalamic dopaminergic neurons
E. Stimulation of anterior pituitary by excessive GnRH secretion

**Rationale:** The patient with Schizophrenia is treated with an anti-psychotic drug. Dopamine normally inhibits prolactin secretion from the anterior pituitary. When dopamine secretion is inhibited by the drug, prolactin secretion increases.

Hyperprolactinemia causes hypogonadism and gynecomastia.

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**Question #:** 22

A 40-year-old woman comes to the physician because of a 2-month history of nervousness, weight loss and inability to tolerate warmth. Her pulse is 105/min, temperature is 37.5°C (99.5°F), blood pressure is 138/74 mm Hg, and respirations are 16/min. Laboratory studies show an elevated serum T4 but a lower serum TSH concentration. Thyroid scan shows decreased iodine uptake by the thyroid gland. Which of the following mechanisms best explains the laboratory findings and clinical condition of this patient?

A. Inhibition of peroxidase enzyme
B. Inhibition pituitary TSH secretion
C. Excessive intake of iodized salt
✓D. Use of exogenous thyroxin (T4)
E. Excessive stimulation of TSH receptors (Grave's disease)

**Rationale:** D is the right answer. Taking exogenous thyroid hormone increases plasma levels of T4 above normal and decreases TSH secretion by negative feedback. This also decreases normal uptake by the thyroid gland. In Graves disease, the same changes in plasma levels of T4 and TSH would be present, but the thyroid gland would be overactive. In fact, goiter is often present in patients with Graves' disease. A lesion in the anterior pituitary that prevents TSH secretion or inhibition of peroxidase enzyme or excessive intake of iodized salt would be associated with low plasma levels of T4.

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**Question #:** 23

A 64-year-old woman comes to the physician because of a 3-week history of excessive thirst and urination, nausea, bone pain and muscle cramps. Her history is significant for breast cancer 2 years ago. Her blood pressure is 150/90 mm Hg and pulse is 78/min. Physical examination shows diffuse tenderness over her abdomen. Laboratory

studies of her blood shows:
Calcium =11.8 mg/dL (normal: 8.4-10.2 mg/dL)
Phosphorous =2.6 mg/dL (normal: 3-4.5 mg/dL)
Total proteins =normal.
The release of which of the following hormones is most likely to be inhibited in this patient?

A. Growth hormone
B. Glucagon
C. Insulin
✓D. Parathyroid hormone
E. Thyroxine

**Rationale:** Blood calcium levels have a direct negative effect on PTH release from the parathyroid glands (negative feedback loop for calcium homeostasis). The elevated calcium levels in this patient would thus inhibit the release of PTH. The other hormones listed as answer options are not affected by calcium levels.

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**Question #:** 24

A 60-year-old man comes to the physician because of a 7-day history of chest discomfort, muscle cramps and numbness of his hands and legs. His blood pressure is 160/108 mm Hg and pulse is 84/min. Physical examination shows no abnormalities. Laboratory studies show:

| Test | Patient value (Reference range) |
|---|---|
| Blood pH | 7.46 (Normal: 7.34-7.40) |
| Serum potassium | 2.4 mEq/L (normal: 3.5−5.0 mEq/L) |
| Plasma aldosterone | abnormally high |
| Plasma renin | Undetectable |

A CT scan of the abdomen shows increased size of the adrenal glands bilaterally. Which of the following additional laboratory findings is most likely in this patient?

✓A. Elevated serum bicarbonate

B. Elevated random blood glucose
C. Low serum sodium
D. Low serum cortisol
E. Low urinary potassium

**Rationale:** A is the correct answer. The clinical presentation of this patient and the diagnostic studies indicate the diagnosis of primary hyperaldosteronism because of bilateral hyperplasia of the zona glomerulosa of the adrenal cortex. High aldosterone produces metabolic alkalosis and thus serum bicarbonate is elevated. C and D are incorrect as the low levels of these reduce blood pressure. This patient has hypertension. Urinary potassium excretion is increased due to the action of the aldosterone on the distal nephron.

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**Question #:** 25

A 32-year-old woman comes to the physician because of a 4-day history of paresthesia, numbness, tingling sensations, and muscle cramps. She had a thyroidectomy a week ago, for a large goiter causing compression of her esophagus resulting in increasing dysphagia over the last 2 years. Physical examination shows positive Chvostek's sign and Trousseau's sign. A diagnosis of hypoparathyroidism is made. She is being administered calcium therapeutically. With respect to regulation of plasma levels of calcium, which of the following is most likely to be considered biologically active?

A. Calcium bound to calmodulin
B. Calcium bound to albumin
C. Calcium phosphate
D. Calcium bound to citrate
✓E. Ionized calcium

**Rationale:** Calcium in plasma or serum exists in three forms or fractions: 1) Protein-bound calcium accounts for approximately one-third of the total serum calcium concentration. Protein-bound calcium cannot diffuse through membranes and thus is not usable by tissues. 2) Ionized or free calcium is the physiologically active form that accounts for 50%–60% of total calcium concentration. 3) Complexed or chelated calcium is bound to phosphate, bicarbonate, sulfate, citrate, and lactate and accounts for ~10% of the total calcium concentration.

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**Question #:** 26

A 26-year-old woman comes to the physician because of a 3-month history of recurrent episodes of headaches and an 8-month history of missing menstrual periods. She says she had regular 28-day cycles before. She also notices a white discharge from her nipples. She is not taking any medications. Her vital signs are within normal limits. Physical examination shows no abnormalities. Pregnancy test is negative. Laboratory studies show normal electrolytes but elevated prolactin level. MRI of brain shows a 10-mm mass in sella turcica. Which of the following additional laboratory findings is most likely in this patient?

A. Increased gonadotropin releasing hormone (GnRH)
B. Elevated serum cortisol
C. High serum growth hormone
D. Abnormal glucose tolerance test
✓E. Low serum FSH and LH

**Rationale:** She has amenorrhea and not pregnant. She has a prolactin secreting tumor. High prolactin suppresses hypothalamic GnRH secretion. Low GnRH causes low LH and FSH and amenorrhea. This is called hypergonadotropic hypogonadism.

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**Question #:** 27

A 29-year-old woman comes to the physician for a follow-up examination. She says she has been urinating hourly accompanied by excessive thirst for the past 2 weeks. She underwent surgical removal of a brain tumor 3 weeks ago. Her vital signs are within normal limits. Physical examination shows dry lips and mucous membranes. Laboratory studies show low serum vasopressin levels. A decreased level of which of the following is most likely in this patient?

A. Serum calcium
B. Random blood glucose
✓C. Urine osmolality
D. Urine volume
E. Serum osmolality

**Rationale:** Increased urine output associated with an excessive thirst and a lab showing low vasopressin goes in favor of the diagnosis of neurogenic diabetes insipidus. Deficiency of vasopressin causes a large volume of urine with low osmolality and low specific gravity.

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**Question #:** 28

A 35-year-old woman comes to the physician because of a 2-month history of progressive fatigue, joint aches, and extreme sensitivity of her skin to the sun. Physical examination shows a butterfly rash on her face and swelling around the joints. Her temperature is 37.7 ºC (99.8 ºF). Her blood test is positive for antinuclear (ANA) antibodies, indicating an autoimmune disorder. She is placed on a high dose of corticosteroids for a prolonged period. Which of the following effects will most likely result from this long-term therapy?

✓A. Osteoporosis
B. Development of male-type hair distribution
C. Deposition of bone and collagen formation
D. Desensitization of beta-adrenergic receptors
E. Hyperplasia of zona fasciculata and reticularis

**Rationale:** Cortisol is prescribed as therapy due to its anti-inflammatory functions. However, another function of cortisol is to cause protein breakdown (proteolysis) in tissues such as muscles and bones. This results in decrease in bone protein and minerals leading to osteoporosis.

Cortisol does not result in increase in body hairs and facial hairs.

Cortisol does not increase, but decreases bone and collagen formation

Cortisol does not affect beta-adrenergic receptors. It does upregulate alpha-1-receptors in the arterioles, thereby increasing sensitivity to the vasoconstrictor norepinephrine

Excess cortisol would results in atrophy and not hyperplasia of the adrenal cortex (it is mainly synthesized in the Z. fasiculata).

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**Question #:** 29

A 26-year-old-woman is brought to the emergency department 10-minutes after she collapses in a park. She was participating in a hunger strike and did not take any food for the past 2 days. Her pulse is 124/min, blood pressure is 106/68 mm Hg and respirations are 18/min. Her blood glucose is 54 mg/dL (Normal: 70-100 mg/dL). She is administered 50% dextrose. In response to hypoglycemia, there is release of a hormone called ghrelin from the stomach. This hormone most likely stimulates which of the following hormones?

A. Somatostatin
✓B. Growth hormone
C. Growth hormone inhibiting hormone
D. Insulin
E. Cortisol

**Rationale:** Besides GHRH, hypoglycemia is a strong trigger for release of GH. It does it indirectly through ghrelin. It does not promote synthesis, instead it is a secretagogue.

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**Question #:** 30

A 25-year-old woman comes to the physician because of a 2-month history of weight loss despite good appetite, nervousness, heat intolerance, and sweaty palms. Her temperature is 37.5°C (99.5°F), pulse is 110/min, and respirations are 18/min. Physical exam shows a diffuse enlargement of her thyroid gland. Diagnosis of hyperthyroidism is made, and she is started on propylthiouracil, a drug that inhibits the action of thyroid peroxidase enzyme in the thyroid gland. Which of the following steps in the synthesis of thyroid hormones is most likely mediated by this enzyme?

A. Synthesis of thyroglobulin
✓B. Oxidation, organification and coupling reactions
C. Iodide uptake by the thyroid gland
D. Conversion of thyroxine (T4) to T3
E. Conversion of thyroxine (T4) to reverse T3

**Rationale:** Propylthiouracil is a drug that inhibits thyroid peroxidase enzyme that inhibits oxidation of Iodide, organification and coupling reactions.